Tingsong Xiao

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LLM Agents | **Post-training** | **World Models** | Full-time availability: 2027

Education

University of Florida Aug 2022 – Dec 2026
Ph.D. in Computer Science | GPA: 3.86/4.00

University of Electronic Science and Technology of China Aug 2017 – Jun 2021
B.Eng. in Computer Science | GPA: 3.92/4.00

Industry Experience

Amazon | Applied Scientist Intern May 2026 – Aug 2026

- On the Amazon **Alexa AI** team, developed the **agentic LLM framework HyperMCTS**, a training-free MCTS planner that **reuses feedback across trajectories** through **hypergraph-guided action priors**. Improved average DeepPlanning accuracy by 2.3–7.3 percentage points over the strongest baseline for each of three backbones, with fewer LLM calls and output tokens than evaluated MCTS-based baselines.
- Developed **AFS** for training-free test-time search with a learned **agentic world model** for **speculative rollouts and action ranking**; executed only selected actions and re-anchored on real observations, matching or exceeding the strongest real-environment search methods with **6–8$\times$ fewer real API calls** on AppWorld and MCPMark Filesystem.
- Built **evaluation pipelines** to analyze task completion, constraint satisfaction, failure modes, and search efficiency, tracking LLM calls, output tokens, and real API usage.

eBay | Applied Researcher Intern May 2025 – Aug 2025

- Developed a **lightweight LLM** for business-critical information extraction, achieving $>$**90% F1 across 10 high-value categories** while reducing inference latency by $>$**5$\times$**, enabling scalable deployment to millions of listings.
- Built an end-to-end **LLM post-training framework** combining **supervised fine-tuning**, multi-strategy **knowledge distillation**, and **reinforcement learning**, improving extraction accuracy, robustness, and deployment efficiency.
- Designed an autonomous **LLM agentic workflow** to synthesize **high-fidelity training data** by converting raw product descriptions into structured aspect-value pairs, reducing reliance on manual labeling and improving robustness on long-tail and rare attribute distributions.

Selected Publications

Peer-reviewed

- Temporally Detailed Hypergraph Neural ODE for Disease Progression Modeling. **ICLR 2026**.
- DecoyDB: A Dataset for Graph Contrastive Learning in Protein-Ligand Binding Affinity Prediction. **NeurIPS 2025**.
- CoastalBench: A Decade-Long High-Resolution Dataset to Emulate Complex Coastal Processes. **ICML 2025**.
- XTSFormer: Cross-Temporal-Scale Transformer for Irregular-Time Event Prediction in Clinical Applications. **AAAI 2025**.
- Spatial-Logic-Aware Weakly Supervised Learning for Flood Mapping on Earth Imagery. **AAAI 2024**.
- A Hierarchical Spatial Transformer for Large Numbers of Point Samples in Continuous Space. **NeurIPS 2023**.

Preprints

- HyperMCTS: Hypergraph-Augmented MCTS for Long-Horizon LLM Agents. **arXiv 2026**.
- EarthSpatialBench: Benchmarking Spatial Reasoning Capabilities of Multimodal LLMs on Earth Imagery. **arXiv 2026**.
- Search in Imagination, Act in Reality: World-Model Test-Time Scaling for Interactive Agents. **Manuscript**.

Open-Source Contribution

Qwen-AgentWorld | **Stars: 1k+**
An open-source **language world model** that simulates interactive environments for training and evaluating **general-purpose AI agents**.

Recognition, Service & Skills

Awards: ACM SIGSPATIAL Best Paper (2023); Gartner Graduate Fellowship (2023, 2025, 2026); AAAI Scholarship (2025); SDM Doctoral Forum Award (2024).
Reviewer: ICLR, ICML, NeurIPS, AAAI; IEEE TPAMI, TKDE, TNNLS.
Skills: Python, C++, PyTorch, Hugging Face, vLLM; supervised fine-tuning, distillation, synthetic data generation, agent planning and evaluation.